

A COUNTEREXAMPLE TO A RAINBOW-DOMINATION IMPLICATION OF ŽEROVNIK

ABSTRACT. Let T be the Tutte–Coxeter graph. We prove

$$(\gamma_{r3}(T), \gamma_{r4}(T), \gamma_{r5}(T)) = (16, 21, 25).$$

Since $|V(T)| = 30$, this gives

$$\gamma_{r5}(T) = \frac{5|V(T)|}{6} \quad \text{but} \quad \gamma_{r4}(T) > \frac{2|V(T)|}{3}.$$

Thus the first implication in Question 2 of Žerovnik fails in general for cubic graphs.

1. INTRODUCTION

For a positive integer k , write $[k] = \{1, \dots, k\}$. A *k-rainbow dominating function* (*k-RDF*) on a graph G is a map $f: V(G) \rightarrow 2^{[k]}$ such that

$$f(v) = \emptyset \implies \bigcup_{u \in N(v)} f(u) = [k].$$

Its weight is $w(f) = \sum_{v \in V(G)} |f(v)|$, and the minimum weight is denoted by $\gamma_{rk}(G)$.

Kuzman proved that every cubic graph G of order N satisfies

$$(1) \quad \gamma_{rk}(G) \geq \left\lceil \frac{kN}{6} \right\rceil, \quad k \leq 6$$

[2, Theorem 1.2]. Following [3], a cubic graph G is *k-rainbow domination regular* when $\gamma_{rk}(G) = k|V(G)|/6$, the extreme case of (1). Writing $N = |V(G)|$, Question 2 of Žerovnik asks whether the reverse chain

$$\gamma_{r5}(G) = \frac{5N}{6} \implies \gamma_{r4}(G) = \frac{2N}{3} \implies \gamma_{r3}(G) = \frac{N}{2}$$

holds for every cubic graph G , that is, whether a counterexample exists [4, Question 2].¹

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¹In [4, Question 2] the chain is printed as $\gamma_{r5}(G) = \frac{5}{3}n \Rightarrow \gamma_{r4}(G) = \frac{4}{3}n \Rightarrow \gamma_{r3}(G) = n$, and n is not redefined at the question; throughout that paper n is the parameter of the generalized Petersen graph $P(n, k)$, which is a graph on $2n$ vertices [4, p. 443]. With $N = |V(G)| = 2n$ the three printed equalities are exactly the equality cases $\gamma_{rk}(G) = kN/6$ of (1) for $k = 5, 4, 3$, which is the chain displayed here. This is also the only non-vacuous reading: assigning $\{1\}$ to every vertex shows $\gamma_{rk}(G) \leq |V(G)|$ for all k and all G , so under the literal reading $n = |V(G)|$ the hypothesis $\gamma_{r5}(G) = \frac{5}{3}n$ could never be satisfied.

Kuzman later noted that the Tutte–Coxeter graph T is not 3-rainbow domination regular, that is, that $\gamma_{r3}(T) \neq 3|V(T)|/6 = 15$ [3, Question (Q6), p. 37]. Since $\gamma_{r5}(T) = 5|V(T)|/6$, as the function f_5 of Section 3 shows, that remark already refutes the implication $\gamma_{r5}(G) = \frac{5N}{6} \Rightarrow \gamma_{r3}(G) = \frac{N}{2}$ obtained by composing the two implications of Question 2. What is added here is the exact middle value $\gamma_{r4}(T) = 21 > 20 = \frac{2|V(T)|}{3}$, which places the failure already in the first implication, together with the exact values of all three parameters.

Theorem 1. *For the Tutte–Coxeter graph T ,*

$$\gamma_{r3}(T) = 16, \quad \gamma_{r4}(T) = 21, \quad \gamma_{r5}(T) = 25.$$

Since $|V(T)| = 30$, Theorem 1 refutes the implication

$$\gamma_{r5}(G) = \frac{5|V(G)|}{6} \implies \gamma_{r4}(G) = \frac{2|V(G)|}{3}$$

for cubic graphs G in general: the graph T satisfies the hypothesis and not the conclusion. The second implication of Question 2 is not addressed by T , which fails its hypothesis; see the Status and scope section.

2. THE DUAD–SYNTHEME MODEL

Let $\Omega = [6]$. A *duad* is a two-element subset of Ω , and a *syntheme* is a partition of Ω into three duads. Let $\mathcal{D}$ and $\mathcal{S}$ be the sets of duads and synthemes. Both have size 15. The graph T is the bipartite incidence graph on $\mathcal{D} \sqcup \mathcal{S}$, with $d \in \mathcal{D}$ adjacent to $s \in \mathcal{S}$ when $d \in s$. Each duad lies in three synthemes, so T is cubic of order 30. The incidence structure is self-dual, and T is connected; indeed T is the Tutte 8-cage, the unique cubic graph of girth 8 on 30 vertices [1]. Being connected and bipartite, T has exactly one bipartition, namely $\{\mathcal{D}, \mathcal{S}\}$.

For either bipartition class X , call $Q \subseteq X$ a *perfect class* if every vertex in the opposite class has exactly one neighbor in Q .

Lemma 2. *At least five perfect classes are required to cover either bipartition class of T .*

Proof. First take $Q \subseteq \mathcal{D}$. Counting incidences with $\mathcal{S}$ gives $3|Q| = 15$, so $|Q| = 5$. No two members of Q are disjoint, since two disjoint duads belong to a common syntheme. Hence Q is a pairwise-intersecting family of five edges of K_6 , and therefore a full star. Indeed, if such a family had no common endpoint, three of its edges would form a triangle, which admits no fourth edge meeting all three.

Conversely, for each $i \in \Omega$, the star

$$\mathcal{D}_i = \{d \in \mathcal{D} : i \in d\}$$

is a perfect class. Thus the perfect classes on $\mathcal{D}$ are precisely the six stars $\mathcal{D}_i$. Four stars leave uncovered the duad joining two unchosen centers, whereas five stars cover $\mathcal{D}$. Self-duality gives the same conclusion on $\mathcal{S}$. $\square$

3. PROOF OF THE THEOREM

We first record the equality structure needed below. Its first assertion is the case $d = 3$, $n = 30$ of [2, Theorem 3.1], restated as [3, Theorem 2.1(ii)]; the short proof is included because we also need the assertion about the individual color supports.

Lemma 3. *Let G be a cubic graph of order 30, let $k < 6$, and let f be a k -RDF of weight $5k$. Then the set of empty and the set of nonempty vertices are independent sets of size 15, so they are the two classes of a bipartition of G . Moreover, each color occurs on exactly five vertices, and every empty vertex has exactly one neighbor carrying each color.*

Proof. Let

$$C = \{v \in V(G) : f(v) \neq \emptyset\}, \quad c = |C|,$$

and let e_C be the number of edges in $G[C]$. Count triples (x, y, i) such that $x \notin C$, $xy \in E(G)$, and $i \in f(y)$. If τ is their number, then rainbow domination gives

$$\tau \geq k(30 - c).$$

On the other hand,

$$\begin{aligned} \tau &= \sum_{y \in C} |f(y)|(3 - \deg_C(y)) \\ &\leq 3w(f) - 2e_C = 15k - 2e_C. \end{aligned}$$

It follows first that $c \geq 15$. The cut from C to its complement has size $3c - 2e_C$ and at most $3(30 - c)$, so

$$e_C \geq 3(c - 15).$$

Consequently,

$$k(30 - c) \leq 15k - 6(c - 15),$$

or equivalently $(6 - k)(c - 15) \leq 0$. Since $k < 6$, we obtain $c = 15$. The preceding inequalities then force $e_C = 0$. All 45 edges incident with C therefore join C to its complement, so the complement is also independent.

For a color i , let $Q_i = \{v : i \in f(v)\}$. Every vertex outside C has a neighbor in Q_i , hence $3|Q_i| \geq 15$. Since

$$\sum_{i=1}^k |Q_i| = w(f) = 5k,$$

we have $|Q_i| = 5$ for every i . Its 15 incident edges meet all 15 vertices outside C , so each such vertex has exactly one neighbor in Q_i . $\square$

Proof of Theorem 1. In each construction below, every syntheme receives the empty label and every duad receives a nonempty label. It is therefore enough to check the syntheme vertices. Nonemptiness of the duad labels is immediate in each of the three cases; note that $\{5, 6\}$ is the only duad disjoint from $[4]$, which is exactly why the definition of f_4 needs one exceptional value.

For five colors, set

$$f_5(d) = d \cap [5], \quad d \in \mathcal{D}.$$

Because each syntheme partitions [6], the union of its three labels is [5]. Each color occurs on five duads, so $w(f_5) = 25$. Together with (1), this gives $\gamma_{r5}(T) = 25$.

For four colors, set

$$f_4(d) = \begin{cases} \{1\}, & d = \{5, 6\}, \\ d \cap [4], & d \neq \{5, 6\}. \end{cases}$$

For every syntheme s ,

$$\bigcup_{d \in s} (d \cap [4]) = [4],$$

and the exceptional label only adds a color. Thus f_4 is a 4-RDF of weight $4 \cdot 5 + 1 = 21$.

For three colors, set

$$f_3(d) = \begin{cases} \{1, 2\}, & d = \{1, 6\}, \\ \{1\}, & 6 \in d \text{ and } 1 \notin d, \\ \{2\}, & 1 \in d \text{ and } 6 \notin d, \\ \{3\}, & d \cap \{1, 6\} = \emptyset. \end{cases}$$

If a syntheme contains $\{1, 6\}$, that duad supplies colors 1, 2 and the other two supply color 3. Otherwise, the duads containing 6 and 1 supply colors 1 and 2, while the third supplies color 3. Hence f_3 is a 3-RDF. Its three color supports have sizes 5, 5, and $\binom{4}{2} = 6$, so $w(f_3) = 16$.

It remains to exclude weights 15 and 20. Suppose that a k -RDF of weight $5k$ exists for $k \in \{3, 4\}$. By Lemma 3, the empty and the nonempty vertices are the two classes of a bipartition of T ; as T is connected, that bipartition is the unique one, so the nonempty vertices form $\mathcal{D}$ or $\mathcal{S}$. Each color support is therefore a perfect class in that bipartition class, and the k supports cover it. This contradicts Lemma 2, since $k < 5$. Therefore

$$\gamma_{r3}(T) \geq 16, \quad \gamma_{r4}(T) \geq 21.$$

The displayed constructions attain these bounds. □

STATUS AND SCOPE

Theorem 1 refutes the first implication of Question 2 of [4] in general for cubic graphs: T is cubic, $\gamma_{r5}(T) = 25 = 5|V(T)|/6$, and $\gamma_{r4}(T) = 21 > 20 = 2|V(T)|/3$. It does not refute the second implication $\gamma_{r4}(G) = \frac{2|V(G)|}{3} \Rightarrow \gamma_{r3}(G) = \frac{|V(G)|}{2}$, whose hypothesis T fails, and we make no claim about that implication. Nothing here characterizes the cubic graphs for which (1) is tight for a given k , and no claim is made about graphs other than T . The chain refuted is the one printed in [4, Question 2] after the substitution

$N = 2n$ recorded in the footnote of Section 1; that substitution is forced, because the printed equalities are the equality cases of (1).

The composed implication was already refutable from the literature. Kuzman's remark that T is not 3-rainbow domination regular [3, Question (Q6), p. 37], together with $\gamma_{r5}(T) = 5|V(T)|/6$, refutes $\gamma_{r5}(G) = \frac{5N}{6} \Rightarrow \gamma_{r3}(G) = \frac{N}{2}$, so a counterexample to the chain read end to end was available before this note. What is added here is the exact middle value $\gamma_{r4}(T) = 21$, which locates the failure already in the first implication, together with the exact values $\gamma_{r3}(T) = 16$ and $\gamma_{r5}(T) = 25$. The first assertion of Lemma 3 is likewise not new: it is the case $d = 3$, $n = 30$ of [2, Theorem 3.1], restated as [3, Theorem 2.1(ii)], and is reproved here only to keep the argument self-contained and to record the statement about individual color supports.

Exact verification. Each of the finite checks enumerated below is recomputable from the data printed in this paper, and each is hand-sized. The only input is the duad–syntheme model of Section 2, which specifies the 15 duads and 15 synthemes of $\Omega = [6]$ completely, together with the three labelings f_5, f_4, f_3 of Section 3. From these one writes out the fifteen synthemes, evaluates each labeling on the three duads of each syntheme and confirms that the union of the three labels is $[5]$, $[4]$ and $[3]$ respectively, that every duad label is nonempty, and that the weights are 25, 21 and 16; Lemma 2 then needs only the fact that the pairwise-intersecting five-member families of duads are the six stars, and Lemma 3 is a counting argument with no computation in it. The checks were re-done independently, from the printed definitions rather than from the first pass, and all 93 of them — the 45 syntheme unions, the 45 nonemptiness conditions and the three weight sums — agreed with the values stated above. The identification of T as the Tutte 8-cage is of a different kind: it rests on the uniqueness of the cubic graph of girth 8 on 30 vertices, which is quoted from [1], is not hand-sized, and is not among those 93 checks; it serves only to name T , and no step of the proof uses the girth.

At this size no program is required, and the 93 checks just listed were carried out by hand. They were nevertheless also run by machine, as a cross-check on that hand work. The incidence structure was rebuilt from the definitions of Section 2 and found to be cubic of order 30, connected and of girth 8; the 45 syntheme unions and the 45 nonemptiness conditions were re-evaluated, with no failing syntheme and no empty duad label for any of f_5, f_4, f_3 ; the perfect classes on the duad side were enumerated exhaustively, returning exactly the six stars $\mathcal{D}_i$ and least cover size five, as in Lemma 2; and the minimum of $\sum_{v,i} x_{v,i}$ over $x \in \{0, 1\}^{V(T) \times [k]}$ subject to

$$\sum_{u \in N(v)} x_{u,i} + \sum_{j \in [k]} x_{v,j} \geq 1 \quad (v \in V(T), i \in [k]),$$

an integer program whose feasible points are exactly the k -RDFs of T under $f(v) = \{i : x_{v,i} = 1\}$ and whose optimum is therefore $\gamma_{rk}(T)$, was computed

to optimality as 16, 21 and 25 for $k = 3, 4, 5$. A verification program accompanies this note and repeats the cross-checks just described, with one difference: it invokes no solver, and obtains 16, 21 and 25 from the displayed labelings and from the lower bounds recalled in the next sentence. No part of the argument rests on them: the upper bounds are the three displayed labelings, and the lower bounds are (1) together with Lemmas 2 and 3.

A search of the rainbow-domination literature, including [2, 3, 4], turned up the question and Kuzman's remark, but no determination of $\gamma_{r_4}(T)$ and no counterexample to the first implication. That search was not exhaustive and has not been through external review, so any earlier resolution we have missed takes precedence over this note.

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